

Loan Approval Prediction

Goal: Build a supervised machine-learning model to predict loan approval while addressing preprocessing, class imbalance, evaluation and deployment trade-offs.

614	Applications
491	Train
123	Test
0.876	Champion ROC-AUC
0.52	Suggested threshold

1. Project Overview & Preprocessing

The dataset contains 614 historical loan applications. An 80/20 stratified split produced 491 training and 123 test records. The notebook excludes **Loan_ID** because it is a unique identifier. Missing values are handled inside the preprocessing pipeline; categorical variables are encoded and numerical variables are scaled. This keeps preprocessing learned from training data rather than leaking test information.

Class imbalance: SMOTE is applied only within the training pipeline. During cross-validation, synthetic minority examples are generated separately inside each training fold, preventing validation contamination.

2. Model Comparison

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.862	0.882	0.872	0.876
Random Forest	0.822	0.871	0.846	0.796
Gradient Boosting	0.820	0.859	0.839	0.774
Decision Tree	0.818	0.847	0.832	0.733

Model trade-off: Logistic Regression is the selected champion for this case study based on test ROC-AUC and provides a comparatively interpretable baseline. Tree-based models can capture nonlinear relationships and interactions, but their test ROC-AUC values were lower in this analysis. All reported metrics treat loan approval (Y) as the positive class.

3. Business Interpretation

- **Credit history:** The notebook's permutation-importance analysis identifies Credit_History as the strongest predictive signal, followed by Property_Area, LoanAmount, Married and Education.
- **Decision quality:** Precision measures the share of predicted approvals that were actually approved; recall measures how many historically approved applications the model recovered.
- **Risk trade-off:** A higher threshold generally increases precision but can reduce recall and therefore reject more potentially good applications. A lower threshold can increase approvals/recall but may increase false approvals and expected credit loss.
- **Governance:** Feature importance represents predictive association, not causation. Sensitive or proxy variables require fairness, legal and policy review before any production lending decision.

4. Cost-Aware Threshold Selection

The notebook does not tune the threshold on the held-out test set. Instead, it generates out-of-fold training probabilities and searches thresholds from 0.10 to 0.90. For the case-study business assumption, a false approval costs 5 units and a false rejection costs 1 unit, while approval recall must remain at least 70%. The resulting candidate threshold is **0.52**.

Policy	Threshold	Precision	Recall	F1	FP	FN
Default	0.50	0.862	0.882	0.872	12	10
Suggested	0.52	0.869	0.859	0.864	11	12

Interpretation: At 0.52, the test-set precision increases from 0.862 to 0.869 and false approvals decrease from 12 to 11, while recall falls from 0.882 to 0.859 and false rejections rise from 10 to 12. This illustrates the approval-risk trade-off rather than proving that 0.52 is universally optimal.

Deployment suggestion: The 0.52 threshold is a candidate for this case study only. A real lender should replace the illustrative 5:1 cost assumption with validated monetary loss estimates, expected loan margin, review costs, risk appetite and approval-volume constraints.

5. Recommended Production Decision Framework

- **High probability:** consider automated approval only after probability calibration, policy checks and validation on recent out-of-time data.
- **Borderline probability:** route to manual review rather than forcing a binary decision; this allows additional documentation and responsible human assessment.
- **Low probability:** consider rejection according to approved lending policy, with appropriate adverse-action reason governance.
- **Continuous monitoring:** track approval rate, realized default/credit outcomes, precision, recall, calibration, data drift, fairness by group and threshold overrides.
- **Model maintenance:** retrain only after data-quality, stability, fairness and compliance checks demonstrate that model performance has changed materially.

6. Conclusion

The project delivers an end-to-end supervised-learning workflow: leakage-safe preprocessing, categorical encoding, numerical scaling, SMOTE-based imbalance handling, comparison of four classification algorithms, metric-based evaluation and cost-aware threshold analysis. Logistic Regression achieved the strongest held-out ROC-AUC (0.876) and was selected as the case-study champion. The 0.52 threshold provides a business-oriented example of balancing false approvals against false rejections, but production deployment requires larger and more recent validation data, calibration, fairness testing, regulatory review and human oversight.

7. Key Skills Demonstrated

Python • Pandas • Scikit-learn • Imbalanced-learn/SMOTE • Pipeline design • Classification • Model evaluation • ROC-AUC • Threshold optimization • Feature importance • Business interpretation

Source: Loan Approval Prediction dataset specified in the Alfido Tech project brief — Kaggle:
<https://www.kaggle.com/datasets/bhanupratapbiswas/loan-approval-prediction-case-study>

Important note: This report reflects the submitted notebook's results and assumptions. The 5:1 false-approval cost is explicitly an illustrative case-study assumption, not a validated lender loss estimate.